Read questions carefully and answer, showing all steps of your solution

Final Answers with no clear solution get ZERO

Work to 5 decimal places in All Questions

Question 1 : Fill in the Blank Questions (with Answers)

1. The _____ error is the difference between the true value and the approximation.
2. The Taylor Series is used to approximate functions using _____ and its derivatives.
3. The Secant Method uses _____ initial approximations to compute the next point.
4. The _____ method requires the derivative of the function to find the next approximation of the root.
5. A necessary condition for applying the Bisection Method is that the function must change _____ over the interval [a, b].

Question 2:

Calculate the true error and relative error for the approximation of $\sqrt{10}$ using the estimated value $\bar{x} = 3.16$.

Question 3:

Use the Newton-Raphson method to approximate a root of $f(x) = \cos(x) - x$ starting with an initial guess $x_0 = 0.5$. Perform 3 iterations.

Iteration	x_n	$f(x_n)$	$f'(x_n)$	x_{n+1}	ε_a (%)
1					
2					
3					

Question 4:

Solving a System of Linear Equations with LU Decomposition

Given the following system of linear equations:

$$x_1 + 2x_2 - x_3 = 2$$
$$2x_1 + x_2 - 2x_3 = 1$$
$$3x_1 - 2x_2 + x_3 = 3$$

x_1	x_2	x_3

Question 5:

Using Gaussian Elimination, find the values of x and y for the system:

$$2x + y = 5$$
$$x + 3y = 7$$

x	y

Question 6 :

Consider the polynomial $P(x) = x^4 - 3x^3 + 20x^2 - 19x + 13$

a) Use Bairstow's Method to find a quadratic factor of the polynomial. Start with an initial guess of $r = -1$ *and* $s = 1$*. Perform 1 iteration.*

b) Use the resulting quadratic factor to find two of the polynomial's roots.

(Q.2).

2. The true error is difference between the true value and the approximation.

(2.) The Taylor Series is used to approximate functions using polynomials and the derivatives.

3. The Secant method uses two intial approximations to compute the next point.

4. The Newton-raphson method requires the derivative of the function to find the next approximation of the root.

5. A necessary condition for applying the Bisection method is that the function must change sign other the interval [a, b].

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[Q.2].

⇒ sol:- T.V = $\sqrt{10}$ = 3.16228 / A.V = 3.16 = $\bar{X}$

relative error

⇒ $\varepsilon_t = \frac{T.V - A.V}{T.V}$ ⇒ $\varepsilon_t = \frac{3.16228 - 3.16}{3.16228}$

$\varepsilon_t = 0.00072$

⇒ True error = T.V − A.V ⇒ True error = 3.16228 − 3.16

⇒ " " = 0.00228

Q.3 sol:- $f(x) = \cos(x) - x$ / $x_0 = 0.5$

⇒ $f'(x) = -\sin(x) - 1$

$\%\varepsilon_{a(1)} = \left|\frac{New.V - old.V}{New.V}\right|$

$\%\varepsilon_a = \left|\frac{0.75522 - 0.5}{0.75522}\right| \times 100$

$\%\varepsilon_{a(1)} = 33.8\%$

⇒[2st iteration]

⇒ $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$ ⇒ $x_1 = 0.5 - \frac{f(0.5)}{f'(0.5)}$

⇒ $x_1 = 0.5 - \frac{0.37758}{-1.47942}$ ⇒ $x_1 = 0.75522$

[2nd iteration]

⇒ $x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$ ⇒ $x_2 = 0.75522 - \frac{f(0.75522)}{f'(0.75522)}$

⇒ $x_2 = 0.75522 - \frac{-0.02709}{-1.68545}$ ⇒ $x_2 = 0.73915$

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$$\%\varepsilon_{a_{(2)}} = \left|\frac{\text{New.V} - \text{old.V}}{\text{New.V}} \times 100\right| \Rightarrow \%\varepsilon_{a_{(2)}} = \left|\frac{0.73915 - 0.75522}{0.73915}\right| 100$$

$$\Rightarrow \%\varepsilon_{a_{(2)}} = 2.17\%$$

$\Rightarrow$ [3rd iteration]

$$X_3 = X_2 - \frac{f(X_2)}{f'(X_2)} \Rightarrow X_3 = 0.73915 - \frac{f(0.73915)}{f'(0.73915)}$$

$$\Rightarrow X_3 = 0.73915 - \frac{-0.00011}{-1.67366} \Rightarrow X_3 = 0.73908$$

$$\Rightarrow \%\varepsilon_{a_{(3)}} = \left|\frac{\text{New.V} - \text{old.V}}{\text{New.V}}\right| \times 100 \Rightarrow \%\varepsilon_{a_{(3)}} = \left|\frac{0.73908 - 0.73915}{0.73908}\right| 100$$

$$\Rightarrow \%\varepsilon_{a_{(3)}} = 0.0095\%$$

i	X_0	$f(X_0)$	$f'(X_0)$	X_{i+1}	$\%\varepsilon_a$
1	0.5	0.37758	-1.47942	0.75522	33.8%
2	0.75522	-0.02709	-1.68545	0.73915	2.17%
3	0.73915	-0.00011	-1.67366	0.73908	0.0095%

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⇒ Ex 2.4

⇒ sol:

$$X_1 + 2X_2 - X_3 = 2$$

$$2X_1 + X_2 - 2X_3 = 1$$

$$3X_1 - 2X_2 + X_3 = 3$$

$$[A]\ [X] = [B]$$

$$\begin{bmatrix} 1 & 2 & -1 \\ 2 & 1 & -2 \\ 3 & -2 & 1 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 2 \\ 2 & 1 & -2 & 1 \\ 3 & -2 & 1 & 3 \end{array}\right] \xrightarrow{-2R_1 + R_2} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 2 \\ 0 & -3 & 0 & -3 \\ 3 & -2 & 1 & 3 \end{array}\right] \xrightarrow{-3R_1 + R_3}$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 2 \\ 0 & -3 & 0 & -3 \\ 0 & -8 & 4 & -3 \end{array}\right] \xrightarrow{-8R_2 + 3R_3} \Rightarrow \left[\begin{array}{ccc|c} 1 & 2 & -1 & 2 \\ 0 & -3 & 0 & -3 \\ 0 & 0 & 12 & 15 \end{array}\right]$$

$$\Rightarrow U = \begin{bmatrix} 1 & 2 & -1 \\ 0 & -3 & 0 \\ 0 & 0 & 12 \end{bmatrix}$$

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$$\Rightarrow L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & \frac{8}{3} & 1 \end{bmatrix}$$

$$\Rightarrow [L][y] = [B]$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & \frac{8}{3} & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$$

① $\Rightarrow y_1 = 2.00000$

② $\Rightarrow (y_1)*2 + y_2 = 1$

$\Rightarrow y_2 = [-3.00000]$

③ $\Rightarrow 3y_1 + \frac{8}{3}y_2 + y_3 = 3$

$\Rightarrow (3*2) + \left(\frac{8}{3} * -3\right) + y_3 = 3$

$\Rightarrow y_3 = 3 - 6 + 8$

$y_3 = 5.00000$

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$$\Rightarrow [U] * [X] = [y]$$

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & -3 & 0 \\ 0 & 0 & 12 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -3 \\ 5 \end{bmatrix}$$

$\Rightarrow$ ~~$X_3 = 12$~~

$\Rightarrow (X_3)12 = 5 \Rightarrow X_3 = \frac{5}{12} = 0.41667$

$\Rightarrow -3X_2 = -3 \Rightarrow \boxed{X_2 = 1.00000}$

$\Rightarrow X_1 + 2X_2 - X_3 = 2$

$\Rightarrow X_1 + (2) - 0.41667 = 2$

$\Rightarrow \boxed{X_1 = 0.41667}$

X_1	X_2	X_3
0.41667	1.00000	0.41667

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(Q2. 5).

⇒ sol:-

$$2x + y = 5$$
$$x + 3y = 7$$

$$\Rightarrow \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \end{bmatrix}$$

$$\Rightarrow \left[\begin{array}{cc|c} 2 & 1 & 5 \\ 1 & 3 & 7 \end{array}\right] -2R_2 + R_1$$

$$\Rightarrow \left[\begin{array}{cc|c} 2 & 1 & 5 \\ 0 & -5 & -9 \end{array}\right] \Rightarrow (0)x_1 + (-5)y = -9$$

$$\Rightarrow y = \frac{-9}{-5} \Rightarrow \boxed{y = 1.8}$$

$$\Rightarrow 2x + y = 5 \Rightarrow 2x + 1.8 = 5$$

$$\Rightarrow 2x = 3.2 \Rightarrow x = \frac{3.2}{2} \Rightarrow \boxed{x = 1.6}$$

x	y
1.6	1.8

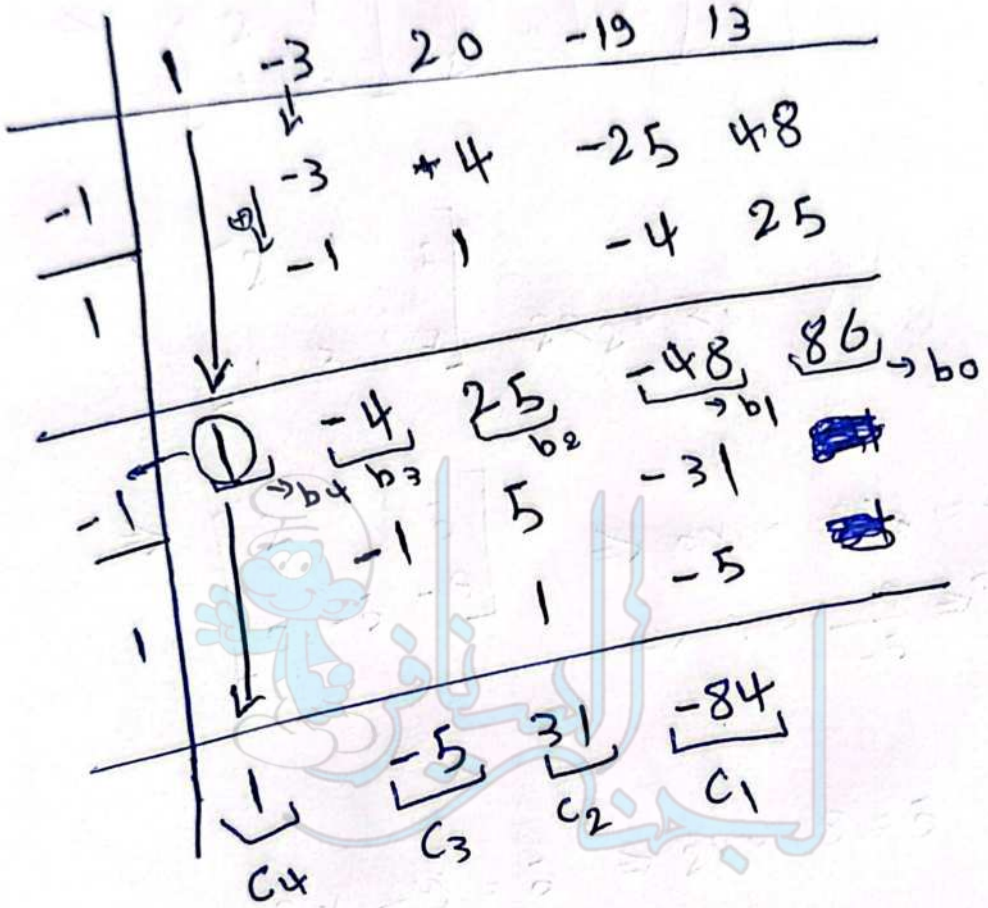
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[Q.6].

⇒ sol:

$$P(x) = x^4 - 3x^3 + 20x^2 - 19x + 13$$

$r_0 = -1$ / $s_0 = 1$



$$\Rightarrow c_1 \Delta r + c_2 \Delta s = -b_0$$
$$c_2 \Delta r + c_3 \Delta s = -b_1$$

$$\Rightarrow \begin{matrix} -84\Delta r + 31\Delta s = -86 \\ 31\Delta r - 5\Delta s = 48 \end{matrix} \Big] \text{من الآلة الحاسبة}$$

$$\begin{bmatrix} \Delta r = 1.95564 \\ \Delta s = 2.5249 \end{bmatrix}$$

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